

DATABASE MANAGEMENT SYSTEMS PROJECT EMPLOYEE PAYROLL AND STATUTORY DEDUCTION MANAGEMENT SYSTEM



PROJECT REPORT

Name: Megha Chayana

Roll No: 25WU0102153

Course: Database Management Systems (DBMS)

Faculty Name: Dr. Kiranmayee

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2. Abstract

Employee payroll processing involves maintaining employee information, department and designation details, attendance, salary components, deductions, monthly payroll and payment records. When such information is maintained manually or in disconnected spreadsheets, it can lead to repeated data entry, inconsistent records and difficulty in retrieving complete payroll information. This project presents the design and implementation of a relational Database Management System for an Employee Payroll and Statutory Deduction Management System.

The system is modelled using an Entity–Relationship diagram and implemented as nine related tables: DEPARTMENT, DESIGNATION, EMPLOYEE, ATTENDANCE, SALARY_COMPONENT, SALARY, DEDUCTION, PAYROLL and PAYMENT. Primary keys identify records uniquely and foreign keys maintain relationships between employees, organizational units, salary information, deductions, payroll and payments.

The database is implemented in MySQL and populated with sample records. SQL queries are used to retrieve employee information, salary and payroll details, attendance, deductions and payment information. A Python Tkinter desktop interface connected through mysql-connector-python provides live viewing, insertion and deletion of records from the MySQL database.

Keywords: DBMS, MySQL, SQL, Python, Tkinter, Payroll Management, Employee Database, Statutory Deduction, ER Diagram, Relational Database

3. Introduction and Problem Statement

3.1 Introduction

An employee payroll system manages information required to calculate, record and process employee compensation. A payroll record is connected to several related entities: an employee belongs to a department and designation, attendance records describe monthly presence, salary records describe earnings, deduction records describe statutory or other deductions, payroll records store the monthly salary summary, and payment records capture how the employee was paid.

A relational DBMS is suitable for this problem because the information is structured and the relationships among employees, departments, designations, salary components, deductions and payments can be represented using primary and foreign keys. The project therefore focuses on designing a consistent relational structure and demonstrating live database operations through a desktop interface.

3.2 Problem Statement

- Employee details may be duplicated when maintained in separate files or registers.
- Department and designation information may become inconsistent when entered repeatedly.
- Attendance, salary, deductions and payroll records need reliable links to the correct employee.
- Statutory deductions such as Provident Fund, Professional Tax and Income Tax need structured storage.
- Payroll and payment information should be retrievable together for an employee.
- Manual searching across multiple records makes payroll-related reporting slower and more error-prone.
- A working database interface is required to demonstrate real insertion, deletion and viewing of records.

The problem addressed by this project is to design and implement a relational database that organizes employee payroll information into connected tables, maintains referential integrity and supports useful SQL queries and live CRUD-oriented demonstrations.

4. Objectives and Scope

4.1 Objectives

- Identify the main entities and relationships required for employee payroll management.
- Design an ER diagram showing primary keys, foreign keys and cardinalities.
- Convert the ER model into a relational schema with nine related tables.
- Apply normalization principles to reduce redundancy and update anomalies.
- Implement the database in MySQL using DDL statements.
- Populate the database with realistic sample employee, salary, deduction, payroll and payment records.
- Write SQL queries for selection, joins, aggregation and employee-specific reporting.
- Develop a Python Tkinter interface connected to MySQL.
- Demonstrate viewing, insertion and deletion of records through the interface.
- Maintain referential integrity using MySQL foreign-key constraints.

4.2 Scope

- Department and designation management.
- Employee registration and employee master data.
- Monthly attendance records.
- Salary components and salary records.
- Statutory and other employee deductions.
- Monthly payroll summary including gross salary, total deductions and net salary.
- Payment records including payment mode, transaction ID, date and status.
- SQL querying and database reporting.
- A Python Tkinter interface for viewing, inserting and deleting records.

4.3 Limitations

The current implementation is a project-level database and desktop demonstration. It does not include a production payroll calculation engine, user authentication/role management, automated tax-rule updates, payslip generation, bank integration or automated statutory filing. These can be considered future enhancements.

5. Software and Hardware Requirements

5.1 Software Requirements

Component	Specification
Operating System	macOS / Mac environment used for development
Database	MySQL 8.x
Database Client	MySQL Workbench / MySQL client
Programming Language	Python 3.x
GUI Library	Tkinter
Database Connector	mysql-connector-python
Code Editor	Visual Studio Code / suitable Python editor
Diagram / Presentation Tool	Diagram and presentation software
Version Control	Git and GitHub

5.2 Hardware Requirements

Component	Requirement
Computer	Mac computer capable of running Python and MySQL
Memory	8 GB RAM or higher recommended
Storage	Sufficient free space for MySQL, Python packages and project files
Display	Standard laptop/desktop display
Network	Internet connection for GitHub access and repository operations
Input Devices	Keyboard and trackpad/mouse

6. ER Diagram

The ER diagram represents the major entities of the Employee Payroll and Statutory Deduction Management System. The central EMPLOYEE entity is connected to organizational information, attendance, salary, deductions and payroll.

SALARY_COMPONENT is referenced by SALARY and DEDUCTION. PAYROLL is connected to PAYMENT, creating the link between payroll processing and the payment transaction.

ER DIAGRAM – EMPLOYEE PAYROLL AND STATUTORY DEDUCTION MANAGEMENT SYSTEM

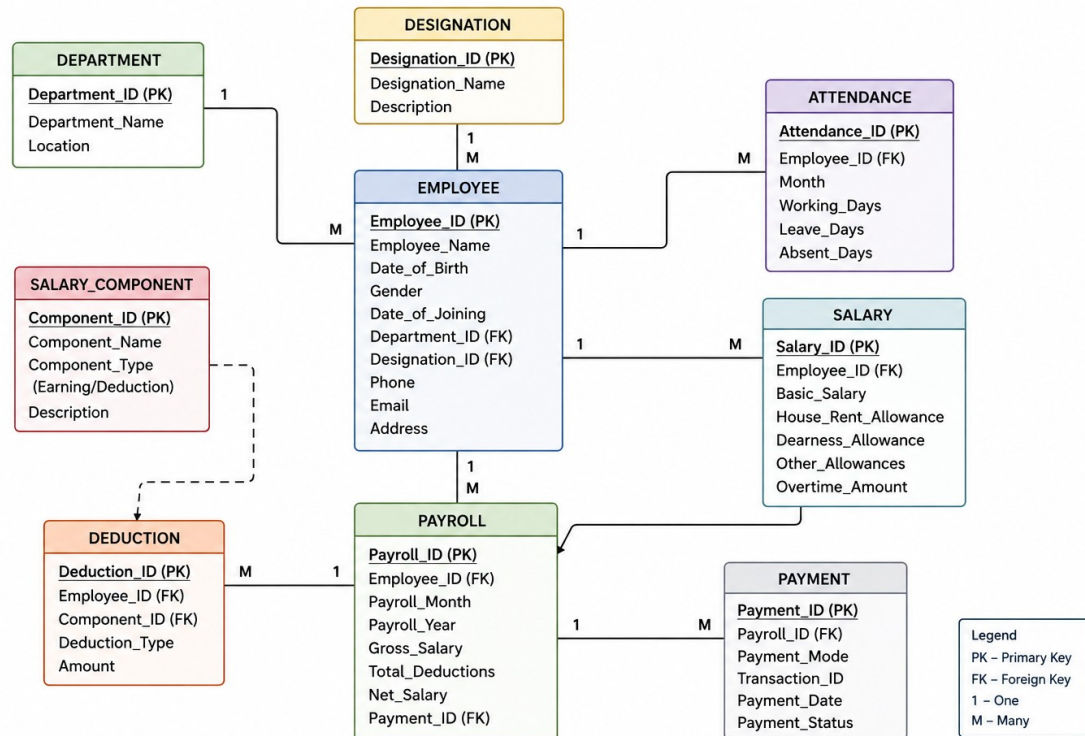


Figure 6.1: ER Diagram of the Employee Payroll and Statutory Deduction Management System

The relationships use one-to-many cardinalities. For example, one department can have many employees, while each employee references one department. Similarly, an employee can have multiple attendance, deduction, salary and payroll-related records. Foreign-key constraints in MySQL enforce these relationships at the database level.

7. Relational Schema and Normalization

7.1 Relational Schema

Relation	Attributes
DEPARTMENT	Department_ID (PK), Department_Name, Location
DESIGNATION	Designation_ID (PK), Designation_Name, Description
EMPLOYEE	Employee_ID (PK), Employee_Name, Date_of_Birth, Gender, Date_of_Joining, Department_ID (FK), Designation_ID (FK), Phone, Email, Address
ATTENDANCE	Attendance_ID (PK), Employee_ID (FK), Month, Working_Days, Leave_Days, Absent_Days
SALARY_COMPONENT	Component_ID (PK), Component_Name, Component_Type, Description
SALARY	Salary_ID (PK), Employee_ID (FK), Component_ID (FK), Basic_Salary, HRA, DA, Other_Allowances, Overtime_Amount
DEDUCTION	Deduction_ID (PK), Employee_ID (FK), Component_ID (FK), Deduction_Type, Amount
PAYROLL	Payroll_ID (PK), Employee_ID (FK), Payroll_Month, Payroll_Year, Gross_Salary, Total_Deductions, Net_Salary, Payment_ID (FK)
PAYMENT	Payment_ID (PK), Payroll_ID (FK), Payment_Mode, Transaction_ID, Payment_Date, Payment_Status

7.2 Key Relationships

Relationship	Meaning
DEPARTMENT → EMPLOYEE	One department can contain many employees.
DESIGNATION → EMPLOYEE	One designation can be assigned to many employees.
EMPLOYEE → ATTENDANCE	An employee can have multiple attendance records.
EMPLOYEE → SALARY	An employee can have salary records.
SALARY_COMPONENT → SALARY	A salary component can be referenced by salary records.

EMPLOYEE → DEDUCTION	An employee can have multiple deduction records.
SALARY_COMPONENT → DEDUCTION	A salary component can represent a deduction.
EMPLOYEE → PAYROLL	An employee can have multiple payroll records.
PAYROLL → PAYMENT	A payroll record can be linked to a payment record.

7.3 Normalization

The schema is designed using normalization principles to keep each fact in an appropriate table and reduce unnecessary repetition.

First Normal Form (1NF). Each table has a primary key and attributes are stored as individual atomic values. Repeating groups are not stored inside a single field.

Second Normal Form (2NF). The tables use single-column primary keys, so non-key attributes do not depend on a partial part of a composite key.

Third Normal Form (3NF). Non-key attributes are kept with the entity they describe. Department details are stored in DEPARTMENT, designation details in DESIGNATION, employee details in EMPLOYEE, and salary/deduction/payment details in their respective tables.

For example, the department name is not repeated in every employee row. EMPLOYEE stores Department_ID and obtains the department information through the DEPARTMENT relationship. Likewise, deduction component information is maintained in SALARY_COMPONENT rather than repeated as a complete component definition in every DEDUCTION row.

8. Data Dictionary

The following data dictionary is based on the implemented SQL schema. PK denotes Primary Key and FK denotes Foreign Key.

8.1 DEPARTMENT

Stores department information such as department name and location.

Column	Data Type	Constraint	Description
Department_ID	INT	PRIMARY KEY	Unique department identifier
Department_Name	VARCHAR(100)	NOT NULL	Department name
Location	VARCHAR(100)	—	Department location

8.2 DESIGNATION

Stores job designations and their descriptions.

Column	Data Type	Constraint	Description
Designation_ID	INT	PRIMARY KEY	Unique designation identifier
Designation_Name	VARCHAR(100)	NOT NULL	Designation name
Description	VARCHAR(255)	—	Description of the designation

8.3 EMPLOYEE

Stores employee personal, joining, contact and organizational information.

Column	Data Type	Constraint	Description
Employee_ID	INT	PRIMARY KEY	Unique employee identifier
Employee_Name	VARCHAR(100)	NOT NULL	Employee name
Date_of_Birth	DATE	—	Employee date of birth
Gender	VARCHAR(10)	—	Employee gender
Date_of_Joining	DATE	—	Joining date
Department_ID	INT	FK → DEPARTMENT	Employee department
Designation_ID	INT	FK → DESIGNATION	Employee designation
Phone	VARCHAR(15)	—	Contact number
Email	VARCHAR(100)	—	Email address
Address	VARCHAR(255)	—	Employee address

8.4 ATTENDANCE

Stores monthly working, leave and absence information for employees.

Column	Data Type	Constraint	Description
Attendance_ID	INT	PRIMARY KEY	Unique attendance record
Employee_ID	INT	FK → EMPLOYEE	Employee reference
Month	VARCHAR(20)	—	Attendance month
Working_Days	INT	—	Number of working days
Leave_Days	INT	—	Approved leave days
Absent_Days	INT	—	Absent days

8.5 SALARY_COMPONENT

Stores earning and deduction components used by the payroll system.

Column	Data Type	Constraint	Description
Component_ID	INT	PRIMARY KEY	Unique component identifier
Component_Name	VARCHAR(100)	NOT NULL	Component name
Component_Type	VARCHAR(50)	—	Earning or Deduction
Description	VARCHAR(255)	—	Component description

8.6 SALARY

Stores employee salary components such as basic salary, HRA, DA and overtime.

Column	Data Type	Constraint	Description
Salary_ID	INT	PRIMARY KEY	Unique salary record
Employee_ID	INT	FK → EMPLOYEE	Employee reference
Component_ID	INT	FK → SALARY_COMPONENT	Salary component reference
Basic_Salary	DECIMAL(10,2)	—	Basic salary
HRA	DECIMAL(10,2)	—	House Rent Allowance
DA	DECIMAL(10,2)	—	Dearness Allowance
Other_Allowances	DECIMAL(10,2)	—	Other allowances
Overtime_Amount	DECIMAL(10,2)	—	Overtime amount

8.7 DEDUCTION

Stores employee statutory/other deductions and their amounts.

Column	Data Type	Constraint	Description
Deduction_ID	INT	PRIMARY KEY	Unique deduction identifier
Employee_ID	INT	FK → EMPLOYEE	Employee reference
Component_ID	INT	FK → SALARY_COMPONENT	Deduction component reference
Deduction_Type	VARCHAR(100)	—	Type of deduction
Amount	DECIMAL(10,2)	—	Deduction amount

8.8 PAYROLL

Stores monthly gross salary, total deductions and net salary for employees.

Column	Data Type	Constraint	Description
Payroll_ID	INT	PRIMARY KEY	Unique payroll identifier
Employee_ID	INT	FK → EMPLOYEE	Employee reference
Payroll_Month	VARCHAR(20)	—	Payroll month
Payroll_Year	INT	—	Payroll year
Gross_Salary	DECIMAL(10,2)	—	Stored gross salary
Total_Deductions	DECIMAL(10,2)	—	Stored total deductions
Net_Salary	DECIMAL(10,2)	—	Stored net salary
Payment_ID	INT	NULL, FK → PAYMENT	Linked payment record

8.9 PAYMENT

Stores payment mode, transaction details, payment date and status.

Column	Data Type	Constraint	Description
Payment_ID	INT	PRIMARY KEY	Unique payment identifier
Payroll_ID	INT	FK → PAYROLL	Payroll reference
Payment_Mode	VARCHAR(50)	—	Payment mode
Transaction_ID	VARCHAR(100)	—	Transaction reference
Payment_Date	DATE	—	Payment date
Payment_Status	VARCHAR(50)	—	Payment status

9. SQL Commands Used (DDL, DML) with Sample Outputs

9.1 Database Creation

```
DROP DATABASE IF EXISTS EmployeePayrollDB;  
CREATE DATABASE EmployeePayrollDB;  
USE EmployeePayrollDB;
```

9.2 DDL – Table Creation

```
CREATE TABLE DEPARTMENT (  
    Department_ID INT PRIMARY KEY,  
    Department_Name VARCHAR(100) NOT NULL,  
    Location VARCHAR(100)  
);  
  
CREATE TABLE EMPLOYEE (  
    Employee_ID INT PRIMARY KEY,  
    Employee_Name VARCHAR(100) NOT NULL,  
    Date_of_Birth DATE,  
    Gender VARCHAR(10),  
    Date_of_Joining DATE,  
    Department_ID INT,  
    Designation_ID INT,  
    Phone VARCHAR(15),  
    Email VARCHAR(100),  
    Address VARCHAR(255),  
    FOREIGN KEY (Department_ID) REFERENCES  
DEPARTMENT(Department_ID),  
    FOREIGN KEY (Designation_ID) REFERENCES  
DESIGNATION(Designation_ID)  
);
```

The remaining tables follow the same relational design. Primary keys provide unique record identification, while foreign keys connect dependent records to their parent records.

9.3 DML – Sample Data Insertion

```
INSERT INTO DEPARTMENT  
(Department_ID, Department_Name, Location)  
VALUES  
(101, 'Human Resources', 'Hyderabad'),  
(102, 'Information Technology', 'Hyderabad'),  
(103, 'Finance', 'Bangalore'),  
(104, 'Marketing', 'Mumbai');  
  
INSERT INTO EMPLOYEE  
(Employee_ID, Employee_Name, Date_of_Birth, Gender,  
    Date_of_Joining, Department_ID, Designation_ID,  
    Phone, Email, Address)  
VALUES
```

```
(1007, 'Vikram Mehta', '1993-12-11', 'Male',
'2018-09-10', 107, 207,
'9876543216', 'vikram@gmail.com', 'Delhi');
```

9.4 Update and Relationship Handling

```
UPDATE PAYROLL
SET Payment_ID = 807
WHERE Payroll_ID = 707;
```

The implemented script first creates and populates PAYROLL and PAYMENT records and then links the corresponding Payment_ID values in PAYROLL. The final schema adds the foreign-key relationship from PAYROLL.Payment_ID to PAYMENT.Payment_ID.

9.5 Viewing All Tables

```
SHOW TABLES;
```

```
SELECT * FROM DEPARTMENT;
SELECT * FROM DESIGNATION;
SELECT * FROM EMPLOYEE;
SELECT * FROM ATTENDANCE;
SELECT * FROM SALARY_COMPONENT;
SELECT * FROM SALARY;
SELECT * FROM DEDUCTION;
SELECT * FROM PAYROLL;
SELECT * FROM PAYMENT;
```

Expected Table Count	Tables
9	DEPARTMENT, DESIGNATION, EMPLOYEE, ATTENDANCE, SALARY_COMPONENT, SALARY, DEDUCTION, PAYROLL, PAYMENT

10. Queries with Outputs

SQL queries were used to demonstrate selection, joins and aggregation. The key query used for Presentation-II was the request to show all available details of Vikram Mehta.

10.1 Presentation-II Query - Vikram Mehta

```
SELECT e.Employee_ID, e.Employee_Name, d.Department_Name,
dg.Designation_Name, a.Month, a.Working_Days, a.Leave_Days,
a.Absent_Days, s.Basic_Salary, s.HRA, s.DA, s.Other_Allowances,
s.Overtime_Amount, p.Gross_Salary, p.Total_Deductions,
p.Net_Salary
FROM EMPLOYEE e
JOIN DEPARTMENT d ON e.Department_ID = d.Department_ID
JOIN DESIGNATION dg ON e.Designation_ID = dg.Designation_ID
LEFT JOIN ATTENDANCE a ON e.Employee_ID = a.Employee_ID
LEFT JOIN SALARY s ON e.Employee_ID = s.Employee_ID
LEFT JOIN PAYROLL p ON e.Employee_ID = p.Employee_ID
WHERE e.Employee_Name = 'Vikram Mehta';
```

Result summary for Vikram Mehta:

ID	Em plo yee	De pt.	De sig nat ion	Mo nth	Wo rk	Le ave	Ab se nt	Ba sic	HR A	DA	Ot her	OT	Gr os s	De d.	Net
1007	Vik ra m Me hta	Sal es	Pro ject Ma nag er	Au gus t	26	0	1	750 00. 00	150 00. 00	750 00. 00	500 00. 00	300 00. 00	100 500 .00	140 00. 00	865 00. 00

The SQL data also records two deductions for Vikram Mehta: Provident Fund of 7,500 and Income Tax of 6,500. His payroll record is Payroll_ID 707 and his payment record is Payment_ID 807, with Bank Transfer, transaction TXN10007, dated 31-Aug-2026 and status Completed.

10.2 Employee and Organizational Details

```
SELECT e.Employee_Name, d.Department_Name, dg.Designation_Name
FROM EMPLOYEE e
JOIN DEPARTMENT d ON e.Department_ID = d.Department_ID
JOIN DESIGNATION dg ON e.Designation_ID = dg.Designation_ID;
```

10.3 Payroll Ordered by Net Salary

```
SELECT e.Employee_Name, p.Payroll_Month, p.Payroll_Year,
p.Gross_Salary, p.Total_Deductions, p.Net_Salary
```

```
FROM EMPLOYEE e
JOIN PAYROLL p ON e.Employee_ID = p.Employee_ID
ORDER BY p.Net_Salary DESC;
```

10.4 Total Deductions by Employee

```
SELECT e.Employee_Name, SUM(d.Amount) AS Total_Deductions
FROM EMPLOYEE e
JOIN DEDUCTION d ON e.Employee_ID = d.Employee_ID
GROUP BY e.Employee_ID, e.Employee_Name
ORDER BY Total_Deductions DESC;
```

10.5 Payment Count by Mode

```
SELECT Payment_Mode, COUNT(*) AS Payment_Count
FROM PAYMENT
GROUP BY Payment_Mode;
```

10.6 Department Listing

```
SELECT Department_ID, Department_Name, Location
FROM DEPARTMENT
ORDER BY Department_ID;
```

10.7 Employees with Absence

```
SELECT e.Employee_Name, a.Month, a.Working_Days,
a.Leave_Days, a.Absent_Days
FROM EMPLOYEE e
JOIN ATTENDANCE a ON e.Employee_ID = a.Employee_ID
WHERE a.Absent_Days > 0;
```

10.8 Salary Comparison

```
SELECT e.Employee_Name, s.Basic_Salary, s.HRA, s.DA,
s.Other_Allowances, s.Overtime_Amount
FROM EMPLOYEE e
JOIN SALARY s ON e.Employee_ID = s.Employee_ID
ORDER BY s.Basic_Salary DESC;
```

11. UI Design and Screenshots

The project includes a desktop GUI developed using Python Tkinter. The interface connects to MySQL database EmployeePayrollDB and displays live records. The main screen lists all nine database tables and provides View/Refresh, Insert and Delete Selected operations.

11.1 Main Database Interface

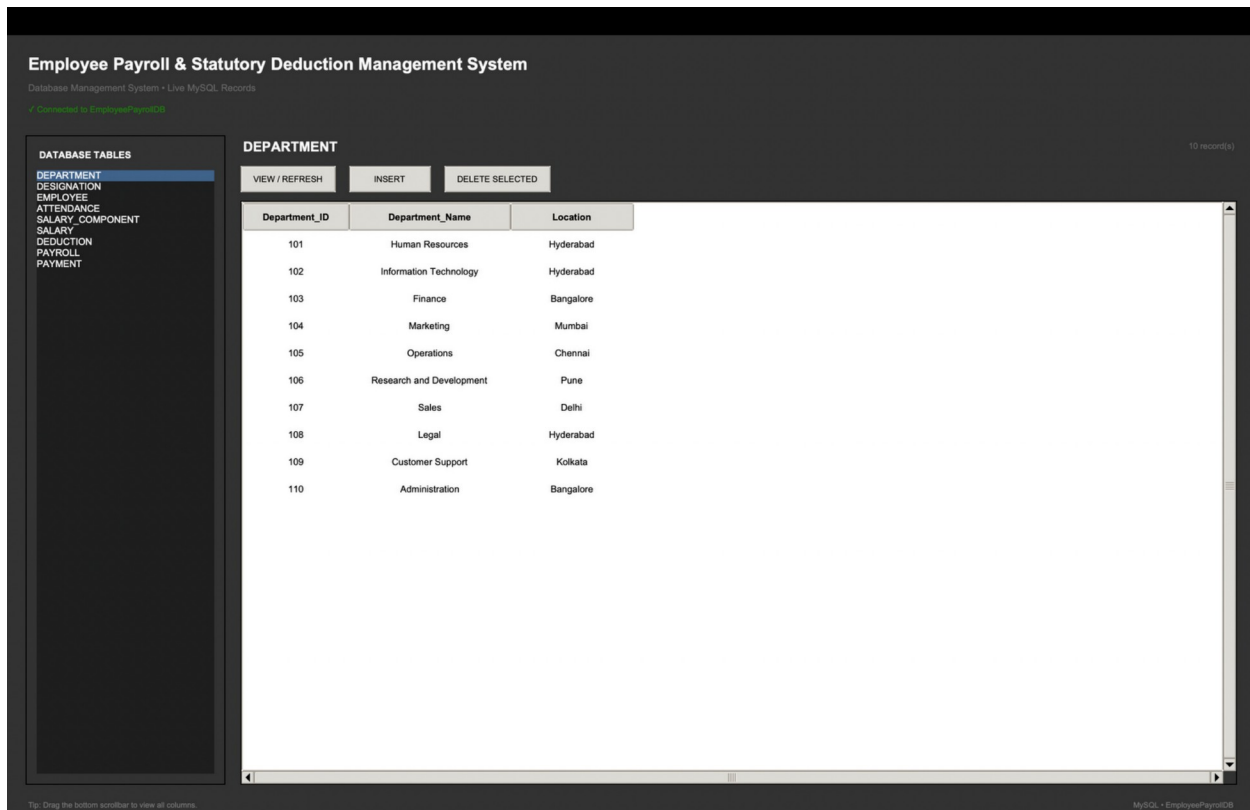


Figure 11.1: Main Tkinter interface showing all nine database tables and live MySQL records

11.2 SALARY Table Before Demonstration Insert

Employee Payroll & Statutory Deduction Management System

Database Management System • Live MySQL Records

✓ Connected to EmployeePayrollDB

10 record(s)

DATABASE TABLES

- DEPARTMENT
- DESIGNATION
- EMPLOYEE
- ATTENDANCE
- SALARY_COMPONENT
- SALARY**
- DEDUCTION
- PAYROLL
- PAYMENT

SALARY

VIEW / REFRESH INSERT DELETE SELECTED

Salary_ID	Employee_ID	Component_ID	Basic_Salary	HRA	DA	Other_Allowances	Overtime_Amount
501	1001	401	50000.00	10000.00	5000.00	3000.00	2000.00
502	1002	401	45000.00	9000.00	4500.00	2500.00	1500.00
503	1003	401	55000.00	11000.00	5500.00	3500.00	2500.00
504	1004	401	40000.00	8000.00	4000.00	2000.00	1000.00
505	1005	401	48000.00	9600.00	4800.00	3000.00	1800.00
506	1006	401	52000.00	10400.00	5200.00	3500.00	2200.00
507	1007	401	75000.00	15000.00	7500.00	5000.00	3000.00
508	1008	401	46000.00	9200.00	4600.00	2800.00	1500.00
509	1009	401	43000.00	8600.00	4300.00	2500.00	1200.00
510	1010	401	68000.00	13600.00	6800.00	4500.00	2500.00

Tip: Drag the bottom scrollbar to view all columns.

MySQL • EmployeePayrollDB

Figure 11.2: SALARY table showing the existing records

11.3 Insert Record Form

Python File Edit Window Help

Employee Payroll & Statutory Deduction Management System

Database Management System • Live MySQL Records

✓ Connected to EmployeePayrollDB

10 record(s)

Insert Record — SALARY

INSERT RECORD

Insert Record — SALARY

Salary_ID

Employee_ID

Component_ID

Basic_Salary

HRA

DA

Other_Allowances

Overtime_Amount

Salary_ID	Employee_ID	Component_ID	Basic_Salary	HRA	DA	Other_Allowances	Overtime_Amount
5000.00			10000.00	5000.00	3000.00	2000.00	
45000.00			9000.00	4500.00	2500.00	1500.00	
55000.00			11000.00	5500.00	3500.00	2500.00	
40000.00			8000.00	4000.00	2000.00	1000.00	
48000.00			9600.00	4800.00	3000.00	1800.00	
52000.00			10400.00	5200.00	3500.00	2200.00	
75000.00			15000.00	7500.00	5000.00	3000.00	
46000.00			9200.00	4600.00	2800.00	1500.00	
43000.00			8600.00	4300.00	2500.00	1200.00	
68000.00			13600.00	6800.00	4500.00	2500.00	

Tip: Drag the bottom scrollbar to view all columns.

MySQL • EmployeePayrollDB

Figure 11.3: Tkinter insert form for the SALARY table

11.4 Live Insert Result

Employee Payroll & Statutory Deduction Management System

Database Management System • Live MySQL Records

✓ Connected to EmployeePayrollDB

DATABASE TABLES

- DEPARTMENT
- DESIGNATION
- EMPLOYEE
- ATTENDANCE
- SALARY_COMPONENT
- SALARY**
- DEDUCTION
- PAYROLL
- PAYMENT

SALARY 11 record(s)

VIEW / REFRESH INSERT DELETE SELECTED

Salary_ID	Employee_ID	Component_ID	Basic_Salary	HRA	DA	Other_Allowances	Overtime_Amount
501	1001	401	50000.00	10000.00	5000.00	3000.00	2000.00
502	1002	401	45000.00	9000.00	4500.00	2500.00	1500.00
503	1003	401	55000.00	11000.00	5500.00	3500.00	2500.00
504	1004	401	40000.00	8000.00	4000.00	2000.00	1000.00
505	1005	401	48000.00	9600.00	4800.00	3000.00	1800.00
506	1006	401	52000.00	10400.00	5200.00	3500.00	2200.00
507	1007	401	75000.00	15000.00	7500.00	5000.00	3000.00
508	1008	401	46000.00	9200.00	4600.00	2800.00	1500.00
509	1009	401	43000.00	8600.00	4300.00	2500.00	1200.00
510	1010	401	68000.00	13600.00	6800.00	4500.00	2500.00
511	1011	401	40000.00	6000.00	4000.00	3000.00	2000.00

All INSERT fields are editable. Foreign-key rules are still enforced by MySQL.

MySQL • EmployeePayrollDB

Figure 11.4: SALARY table after insertion of a temporary demonstration record

11.5 Delete Confirmation

Employee Payroll & Statutory Deduction Management System

Database Management System • Live MySQL Records

✓ Connected to EmployeePayrollDB

DATABASE TABLES

- DEPARTMENT
- DESIGNATION
- EMPLOYEE
- ATTENDANCE
- SALARY_COMPONENT
- SALARY**
- DEDUCTION
- PAYROLL
- PAYMENT

SALARY 11 record(s)

VIEW / REFRESH INSERT DELETE SELECTED

Salary_ID	Employee_ID	Component_ID	Basic_Salary	HRA	DA	Other_Allowances	Overtime_Amount
501	1001	401	50000.00	10000.00	5000.00	3000.00	2000.00
502	1002	401	45000.00	9000.00	4500.00	2500.00	1500.00
503	1003	401	55000.00	11000.00	5500.00	3500.00	2500.00
504	1004	401	40000.00	8000.00	4000.00	2000.00	1000.00
505	1005	401	48000.00	9600.00	4800.00	3000.00	1800.00
506	1006	401	52000.00	10400.00	5200.00	3500.00	2200.00
507	1007	401	75000.00	15000.00	7500.00	5000.00	3000.00
508	1008	401	46000.00	9200.00	4600.00	2800.00	1500.00
509	1009	401	43000.00	8600.00	4300.00	2500.00	1200.00
510	1010	401	68000.00	13600.00	6800.00	4500.00	2500.00
511	1011	401	40000.00	6000.00	4000.00	3000.00	2000.00

Delete this record?
Salary_ID = 511

No Yes

All INSERT fields are editable. Foreign-key rules are still enforced by MySQL.

MySQL • EmployeePayrollDB

Figure 11.5: Delete confirmation for the selected demonstration record

The screenshots demonstrate that the interface reads the current database state and that insert and delete actions are reflected in the table view. The demonstration record is separate from the main sample dataset and was used to show live CRUD behaviour.

12. Implementation Details

12.1 Technology Stack

Layer	Technology
Database	MySQL
Query Language	SQL – DDL, DML and DQL
Application Language	Python 3.x
User Interface	Tkinter desktop GUI
Database Connector	mysql-connector-python
Database Client	MySQL Workbench / MySQL client
Version Control	Git and GitHub

12.2 Database Connectivity

The Python application uses mysql-connector-python to establish a connection to the local MySQL server and the EmployeePayrollDB database. SQL statements are executed through the connection, and the returned rows are displayed in the Tkinter table view. The interface therefore works against the live database rather than a static copy of the records.

12.3 Application Features

- Displays all nine database tables.
- Loads and refreshes live MySQL records.
- Provides an insert form based on the selected table.
- Allows the user to select a row and delete it after confirmation.
- Displays database errors when a database constraint prevents an operation.
- Keeps MySQL primary-key and foreign-key rules active during UI operations.
- Demonstrates the database connection status and current record count.

12.4 Representative Connection Code

```
import mysql.connector

connection = mysql.connector.connect(
    host="localhost",
    user="payroll_app",
    password="PayrollDemo@2026!",
    database="EmployeePayrollDB"
)

cursor = connection.cursor()
cursor.execute("SELECT * FROM EMPLOYEE")
rows = cursor.fetchall()
```

The application account is used for the project interface so that the UI can connect to the EmployeePayrollDB database without changing the MySQL root password.

12.5 Insert and Delete Flow

1. Select a table from the table list.
2. Click INSERT and enter values for the table columns.
3. The application executes an INSERT statement.
4. The interface refreshes the selected table and displays the new row.
5. For deletion, select a row and click DELETE SELECTED.
6. The application asks for confirmation before executing DELETE.
7. After deletion, the table is refreshed to show the live database state.

13. Testing

Testing focused on database integrity and the live UI operations required for the project. The database schema contains primary-key, NOT NULL and foreign-key constraints. These constraints should reject invalid records and protect parent-child relationships.

ID	Test Case	Input / Action	Expected / Observed Behaviour	Status
T1	Duplicate primary key	Insert a DEPARTMENT row using an existing Department_ID	Operation rejected by PRIMARY KEY	Pass
T2	Mandatory field	Insert DEPARTMENT with Department_Name = NULL	Operation rejected by NOT NULL	Pass
T3	Invalid department FK	Insert EMPLOYEE with Department_ID that does not exist	Operation rejected by FOREIGN KEY	Pass
T4	Invalid designation FK	Insert EMPLOYEE with non-existent Designation_ID	Operation rejected by FOREIGN KEY	Pass
T5	Invalid employee FK	Insert ATTENDANCE for a non-existent Employee_ID	Operation rejected by FOREIGN KEY	Pass
T6	Invalid component FK	Insert SALARY using non-existent Component_ID	Operation rejected by FOREIGN KEY	Pass
T7	Invalid payroll employee	Insert PAYROLL for a non-existent Employee_ID	Operation rejected by FOREIGN KEY	Pass

T8	Invalid payment payroll	Insert PAYMENT with non-existent Payroll_ID	Operation rejected by FOREIGN KEY	Pass
T9	Delete referenced parent	Delete an employee that has dependent records	Referential integrity prevents unsafe deletion	Pass
T10	View all tables	Select each of the nine tables in the UI	Current MySQL records are displayed	Pass
T11	Insert record	Insert temporary SALARY demonstration row	Row appears in live table view	Pass
T12	Delete record	Select temporary row and confirm deletion	Row is removed and table refreshes	Pass
T13	Vikram query	Run the Presentation-II employee query	Vikram Mehta details are returned	Pass

Note: The database-level test outcomes for constraint cases are based on the implemented primary-key, NOT NULL and foreign-key definitions. The UI insert/delete cases are demonstrated by the provided screenshots, including the temporary SALARY record and its deletion confirmation.

14. Conclusion and Future Enhancements

14.1 Conclusion

The Employee Payroll and Statutory Deduction Management System successfully demonstrates the design and implementation of a relational DBMS for employee payroll information. The project starts with an ER model and maps it to nine related tables covering departments, designations, employees, attendance, salary components, salaries, deductions, payroll and payments.

The MySQL implementation uses primary keys and foreign keys to maintain record identity and referential integrity. Sample data provides sufficient records for SQL querying and demonstration. The Presentation-II employee query shows how multiple related tables can be joined to retrieve a complete employee payroll view. The Python Tkinter application connects to the live database and demonstrates viewing, insertion and deletion of records.

14.2 Future Enhancements

- Add user authentication and role-based access for HR, payroll administrators and managers.
- Add Edit/Update functionality to complement the current view, insert and delete operations.
- Add search and filtering by employee name, department, designation, payroll month and payment status.
- Generate downloadable employee payslips and payroll reports.
- Automate payroll calculations from salary components, attendance and deduction rules.
- Add configurable statutory deduction rules and tax slabs.
- Create stored procedures and views for monthly payroll summaries and management reports.
- Add audit logging so that important database changes can be traced.
- Add automated backups and recovery procedures.
- Extend the desktop interface into a web or mobile application for wider access.

15. References

- [1] R. Elmasri and S. B. Navathe, Fundamentals of Database Systems, 7th ed., Pearson, 2016.
- [2] A. Silberschatz, H. F. Korth and S. Sudarshan, Database System Concepts, 7th ed., McGraw-Hill, 2019.
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- [4] Oracle Corporation, MySQL 8.0 Reference Manual.
- [5] P. P. Chen, "The Entity-Relationship Model – Toward a Unified View of Data," ACM Transactions on Database Systems, vol. 1, no. 1, 1976.
- [6] Python Software Foundation, Python Documentation.
- [7] TkDocs, Tkinter documentation and reference material.

Project-specific sources: the implemented EmployeePayrollDB SQL script, ER diagram, Tkinter UI screenshots and GitHub repository supplied for this project.

16. Appendix: GitHub Repository Link

The complete project repository is maintained on GitHub. It contains the project presentations, SQL implementation, ER diagram, UI/project files and report materials.

<https://github.com/meghachayana/DBMS-Course-Project>

Repository: DBMS-Course-Project

Student: Megha Chayana

Roll No: 25WU0102153

Course: Database Management Systems

The repository should be treated as the version-controlled source for the final project deliverables.